Dynamic Replacement Model: Fit Summary across 4p, 8p, and 8p+FE Specifications

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1 Introduction

This report summarizes the in-sample fit of the dynamic structural replacement model under four specifications, each estimated by Aguirregabiria-Mira NPL on the observed UST panel (N = 2,282,735 tank-year observations):

1. **4-parameter** — a single $(\kappa, K, \gamma_{\text{price}}, \gamma_{\text{risk}})$ block, pooled across wall and regime.
2. **8-parameter** — the same primitives split by wall type (SW/DW) and insurance regime (FF/RB): 4 cost parameters and 4 price/risk slopes.
3. **8-parameter + FE (M-only, joint)** — the 8p structural specification augmented with 17 state-level intercepts α_g entering the *Maintain* flow utility only. Estimated jointly (25 parameters in `optim`).
4. **8-parameter + stayFE (profile)** — the same 8 structural parameters with 17 state-level intercepts entering both the *Maintain* and *Replace* flow utilities (the "stay-in-business" actions), with the α_g *profiled out* via a 1-D Newton FOC inside each likelihood evaluation (so `optim` searches only the 8 structural dimensions). The exit utility is unchanged across all four specifications and serves as the outside option.

All four specifications use the *Semantic 2* convention for the α_g: the FE enter the measurement equation (the likelihood) only, not the agent's value function in counterfactual re-solves. Texas is the baseline ($\alpha_{\text{TX}} \equiv 0$).

The report is organized for direct visual and statistical comparison: Section 2 gives the flow-utility specifications; Section 3 reports the four parameter tables and the FE alphas; Section 4 reports the four-way information-criteria and likelihood-ratio table plus an aggregate fit-metric comparison; Section 5 reports model-implied vs. empirical action shares at the (wall × regime × age bin) cell level for each specification, with figures produced by a single shared plotting routine to guarantee structural parity. Per-cell numerical fit tables are in Section 6.

Artifacts are produced by `Code/Dynamic_Model/04k_Fit_Report_Artifacts.R` (specs 1–3) and `Code/Dynamic_Model/04l_8paramFE_Profile_Estimation.R` (spec 4) from saved fit objects under `Output/Estimation_Results/`. No estimation is re-run in the production of this report.

2 Model Specifications

All four specifications share the same dynamic discrete-choice (DCC) structure with three actions $a_t \in \{M, E, R\}$ (Maintain, Exit, Replace), discount factor $\beta = 0.95$, T1EV(0) shocks with scale $\sigma_2 = 1$, and a state space of $S = 32$ cells indexed by tank age bin $A_{\text{bin}} \in \{1, \dots, 8\}$ (5-year bins), wall type $w \in \{\text{SW}, \text{DW}\}$, and insurance regime $\rho \in \{\text{FF}, \text{RB}\}$. The exit action is absorbing (`F_exit` does not enter the Bellman operator), so u_E is the firm's terminal outside-option payoff. All cost parameters κ and K are reported in dollar units (one model unit = \$10,000/yr/facility).

2.1 4-parameter flow utilities

A single $(\kappa, K, \gamma_p, \gamma_r)$ vector applies to every state cell. The flow utilities at state s are:

$$
\begin{aligned}
u_M(s;\theta_{4p}) &= 1 + \gamma_{\text{price}} \cdot P(s) - \gamma_{\text{risk}} \cdot h(s) \\
u_E(s;\theta_{4p}) &= \kappa \\
u_R(s;\theta_{4p}) &= -K = -\exp(K_{\log})
\end{aligned}
$$

where $P(s)$ is the per-period insurance premium in cell s, $h(s)$ is the cell's hazard-loss product (probability of release $\times$ expected cleanup cost), and the leading 1 in u_M is the normalized annual net revenue $R = \$10{,}000/\text{yr}$ collected each period the facility operates. Parameter vector: $\theta_{4p} = (\kappa_{\text{exit}}, K_{\log}, \gamma_{\text{price}}, \gamma_{\text{risk}})$, 4 entries.

2.2 8-parameter flow utilities

Cost parameters split by wall, price/risk semi-elasticities split by regime:

$$
\begin{aligned}
u_M(s;\theta_{8p}) &= 1 + \gamma_{\text{price},\rho(s)} \cdot P(s) - \gamma_{\text{risk},\rho(s)} \cdot h(s) \\
u_E(s;\theta_{8p}) &= \kappa_{w(s)} \\
u_R(s;\theta_{8p}) &= -K_{w(s)} = -\exp(K_{\log,w(s)})
\end{aligned}
$$

with $w(s) \in \{\text{SW}, \text{DW}\}$ and $\rho(s) \in \{\text{FF}, \text{RB}\}$ read off the state cell. Parameter vector: $\theta_{8p} = (\kappa_{\text{SW}}, \kappa_{\text{DW}}, K_{\log,\text{SW}}, K_{\log,\text{DW}}, \gamma_{\text{price,FF}}, \gamma_{\text{price,RB}}, \gamma_{\text{risk,FF}}, \gamma_{\text{risk,RB}})$, 8 entries.

Sign conventions: $\gamma_{\text{price}} < 0$ in the parameter (premium *reduces* maintain utility); $\gamma_{\text{risk}} > 0$ in the parameter but enters with a minus sign in the equation, so higher hazard also *reduces* maintain utility.

2.3 8-parameter + FE (M-only, joint)

Adds an additive group intercept α_g to the Maintain flow only, indexed by the firm's state of operation $g \in \{0 = \text{TX}, 1 = \text{AR}, \ldots, 17 = \text{VA}\}$:

$$
\begin{aligned}
u_M(s, g;\theta_{8p}, \alpha_g) &= v_M(s;\theta_{8p}) + \alpha_g \\
u_E(s;\theta_{8p}) &= v_E(s;\theta_{8p}) \\
u_R(s;\theta_{8p}) &= v_R(s;\theta_{8p})
\end{aligned}
$$

where $v_j(s;\theta)$ are the NPL choice-specific values from the existing 8p model (i.e., the same flow utilities as Section 2.2 plus continuation values where applicable). Texas is the baseline ($\alpha_0 \equiv 0$). The 17 control-state alphas are estimated *jointly* with θ_{8p} in a 25-parameter `optim` call. Parameter vector: θ_{8p} plus $(\alpha_{\text{AR}}, \alpha_{\text{CO}}, \ldots, \alpha_{\text{VA}})$, $8 + 17 = 25$ entries.

2.4 8-parameter + stayFE (profile)

The state intercept α_g shifts both *Maintain* and *Replace* utilities — the two "stay-in-business" actions — leaving the exit (outside-option) utility untouched:

Table 1: 4-parameter NPL estimates: pooled κ, K, γ_{price}, γ_{risk}.

	Observed (headline) TX 2006+, controls 1999+	Extended (2000+) TX 2000+, controls 2000+
κ_{exit}	\$261,222	\$257,702
	(\$306)	(\$310)
	[\$260,621, \$261,822]	[\$257,094, \$258,310]
K	\$51,779	\$52,658
	(\$142)	(\$146)
	[\$51,501, \$52,057]	[\$52,371, \$52,945]
γ_{price}	-0.293	-1.142
	(0.049)	(0.048)
	[-0.389, -0.197]	[-1.236, -1.048]
γ_{risk}	0.164	0.153
	(0.003)	(0.003)
	[0.158, 0.170]	[0.147, 0.159]
$\log L$	-268,481	-247,614
N obs	2,282,735	2,300,780

Point estimates with AM (2002) profile-likelihood standard errors in parentheses and 95% CIs in brackets. Bootstrap SEs deferred.

$$
\begin{aligned}
u_M(s, g; \theta_{8p}, \alpha_g) &= v_M(s; \theta_{8p}) + \alpha_g \\
u_E(s; \theta_{8p}) &= v_E(s; \theta_{8p}) \\
u_R(s, g; \theta_{8p}, \alpha_g) &= v_R(s; \theta_{8p}) + \alpha_g
\end{aligned}
$$

Crucially, α_g is *profiled out* of the likelihood: for each θ_{8p} candidate the optimizer evaluates, the 17 control-state α_g values are solved analytically from a 1-D Newton FOC enforcing

$$
\sum_s \Big[n_M(s,g) + n_R(s,g) - n_{\text{total}}(s,g) \cdot \hat{P}_{\text{stay}}(s, g; \theta_{8p}, \alpha_g) \Big] = 0
$$

(i.e., the model-implied total "stay" count in group g equals the empirical count). `optim` therefore searches only the 8 structural dimensions. Texas is the baseline ($\alpha_0 \equiv 0$). At convergence, the $\alpha_g(\hat{\theta})$ values are recovered post-hoc. See `Reports/Teaching_Notes_Dynamic_Models.md` §1 for the full derivation, literature, and standard-error treatment. Parameter vector: 8 structural plus 17 profiled-out alphas; effective $k = 25$ for information-criteria purposes.

3 Parameter Estimates

3.1 4-parameter specification

Table 1 reports point estimates with AM (2002) profile-likelihood standard errors and 95% Wald confidence intervals.

Table 2: 8-parameter NPL estimates: κ and K split by wall (SW/DW); γ_{price} and γ_{risk} split by regime (FF/RB).

	Observed (headline) TX 2006+, controls 1999+
κ_{SW}	\$236,795
	(\$716)
	[\$235,392, \$238,198]
κ_{DW}	\$211,775
	(\$856)
	[\$210,098, \$213,452]
K_{SW}	\$58,085
	(\$212)
	[\$57,670, \$58,501]
K_{DW}	\$43,301
	(\$244)
	[\$42,823, \$43,779]
$\gamma_{\text{price,FF}}$	-14.382
	(0.293)
	[-14.957, -13.808]
$\gamma_{\text{price,RB}}$	-7.485
	(0.151)
	[-7.781, -7.188]
$\gamma_{\text{risk,FF}}$	0.065
	(0.005)
	[0.055, 0.075]
$\gamma_{\text{risk,RB}}$	-0.083
	(0.009)
	[-0.100, -0.066]
$\log L$	-261,770
N obs	2,282,735

Point estimates with AM (2002) profile-likelihood SEs in parentheses and 95% CIs in brackets. MC robustness battery on 8-param spec deferred.

3.2 8-parameter specification

Table 2 splits κ and K by wall type (SW/DW) and the price/risk slopes by insurance regime (FF/RB).

3.3 8-parameter + FE (M-only, joint)

Table 3 reports the 8-structural-parameter block from the joint 25-dim (θ, α) estimation with α_g entering u_M only. The 17 state fixed effects are reported separately in Table 4.

3.3.1 State fixed-effect coefficients

The state fixed effects α_g enter the measurement equation (the likelihood) only — they shift the model-implied probability of *Maintain* in state g relative to the TX baseline. They do **not** enter the agent's equilibrium value function during counterfactual re-solves, per the "Semantic 2" convention adopted in this codebase.

Note on extreme estimates. The Missouri ($\hat{\alpha}_{\text{MO}} = 8.18$, SE 0.77) and South Dakota ($\hat{\alpha}_{\text{SD}} = 2.27$, SE 0.08) coefficients are substantially larger than the other 15 control-state alphas, which span roughly $[-1.4, +0.8]$. Both are statistically distinguishable from zero and reflect near-zero observed *Exit*

Table 3: 8-parameter + FE NPL estimates: structural parameters from the joint (θ, α) fit. The 17 state fixed effects α_g are reported separately in Table 4 (omitted here to avoid duplication).

Parameter	Estimate	(SE)	95% CI
κ_{SW}	\$246,706	(\$811)	[\$245,118, \$248,295]
κ_{DW}	\$219,264	(\$925)	[\$217,451, \$221,078]
K_{SW}	\$55,331	(\$362)	[\$54,622, \$56,041]
K_{DW}	\$43,051	(\$359)	[\$42,347, \$43,756]
$\gamma_{\text{price,FF}}$	-11.088	(0.343)	[-11.760, -10.416]
$\gamma_{\text{price,RB}}$	-6.506	(0.163)	[-6.826, -6.185]
$\gamma_{\text{risk,FF}}$	0.088	(0.005)	[0.078, 0.098]
$\gamma_{\text{risk,RB}}$	-0.120	(0.009)	[-0.137, -0.102]
$\log L$		-251,613	
N obs		2,282,735	

Observed sample. TX FE fixed at 0. 17 control-state maintain-shifters estimated.

Table 4: State fixed-effect coefficients $\hat{\alpha}_g$ from the 8p+FE specification. Texas is the baseline ($\alpha_{\text{TX}} \equiv 0$); standard errors in parentheses.

State	$\hat{\alpha}_g$ (SE)
TX	0 (baseline)
AR	0.095 (0.040)
CO	0.064 (0.042)
ID	-0.114 (0.049)
KS	0.478 (0.043)
KY	-0.124 (0.038)
LA	-0.449 (0.037)
MA	-0.415 (0.039)
MD	-0.360 (0.037)
ME	-1.393 (0.042)
MN	0.290 (0.040)
MO	8.181 (0.774)
NC	0.572 (0.038)
OH	0.064 (0.037)
OK	0.151 (0.039)
SD	2.269 (0.083)
TN	0.187 (0.038)
VA	0.803 (0.039)

Texas is the baseline (alpha set to 0). Other states are estimated jointly with structural theta in the 25-parameter 8p+FE specification.

Table 5: 8-parameter + stayFE (profile) estimates: structural parameters only. The 17 state FE coefficients α_g (profiled out at $\hat{\theta}$) are reported separately in Table 6.

Parameter	Estimate	(SE)	95% CI
κ_{SW}	\$239,188	(\$733)	[\$237,751, \$240,624]
κ_{DW}	\$214,380	(\$875)	[\$212,665, \$216,094]
K_{SW}	\$57,186	(\$210)	[\$56,775, \$57,596]
K_{DW}	\$42,554	(\$241)	[\$42,082, \$43,026]
$\gamma_{\text{price,FF}}$	-10.281	(0.286)	[-10.841, -9.721]
$\gamma_{\text{price,RB}}$	-7.045	(0.154)	[-7.347, -6.743]
$\gamma_{\text{risk,FF}}$	0.058	(0.005)	[0.048, 0.067]
$\gamma_{\text{risk,RB}}$	-0.077	(0.009)	[-0.094, -0.059]
$\log L$			-252,365
N obs			2,282,735

8-parameter structural fit with state FE on {Maintain, Replace} profiled out via 1-D FOC at each θ-evaluation. SEs are profile-likelihood AM (2002) form (Schur complement of joint 25-dim Hessian). State-FE coefficients α_g are reported in Table 04l_FE_Alphas_profile_Compact.tex.

and *Replace* activity in those state-year cells relative to the TX baseline; they are not numerical artifacts.

3.4 8-parameter + stayFE (profile)

Table 5 reports the 8 structural parameters from the profile-likelihood fit. The 17 state intercepts α_g are recovered post-hoc at $\hat{\theta}$ and reported in Table 6.

3.4.1 State fixed-effect coefficients (profile spec)

Note on the MO boundary. The Missouri alpha hits the ± 20 Newton bound ($\hat{\alpha}_{\text{MO}} = 20.0$, SE 9.46) because the group's empirical Exit-share is essentially zero in the observed window; the profile FOC has no interior solution and the solver correctly pins the value at the bound with an inflated SE that honestly reports the local non-identification. This is an improvement over the M-only joint spec where the same data pathology surfaced as a spuriously-tight $\hat{\alpha}_{\text{MO}} = 8.18$ (SE 0.77).

4 Model Comparison

Table 7 reports log-likelihood, information criteria, and nested likelihood-ratio tests across the four specifications. All four are fit on the same observed sample, so the comparison is direct.

All three nested restrictions are rejected at any conventional level. AIC and BIC agree on the ranking 4p $\succ$ 8p $\succ$ 8p+stayFE (profile) $\succ$ 8p+FE (M-only, joint) — the joint-M-only fit is best in absolute likelihood terms. However, the two FE specifications are *non-nested* to each other: they differ in where α_g enters the flow utility, not in parameter count. The profile-stayFE specification has the cleaner economic interpretation (α_g shifts state-specific stay-vs-exit propensity) at a cost of ≈ 750 log-likelihood units relative to the M-only joint fit.

4.1 Aggregate fit-metric comparison

Table 8 collapses the per-(wall, regime, action) weighted-RMSE numbers from each model into a per-action summary weighted by total cell sample size. Lower is better in every column.

Takeaway: the 8p specifications all reduce overall weighted RMSE by roughly half relative to 4p; among the 8p variants, the M-only-joint spec achieves the lowest overall RMSE, with the stayFE-profile spec a close second. The 4p model's poor fit is concentrated in *Exit* and *Maintain* (where

Table 6: State fixed-effect coefficients $\hat{\alpha}_g(\hat{\theta})$ from the stayFE-profile specification, recovered post-convergence from the profile FOC. Texas is the baseline ($\alpha_{\mathrm{TX}} \equiv 0$); standard errors in parentheses are from the joint 25-dim Hessian sub-block. Several SE values report as 0.000 due to finite-difference precision in `numDeriv::hessian`; these are not exact zeros but values below the 3-decimal display threshold.

State	$\hat{\alpha}_g$ (SE)
TX	0 (baseline)
AR	-0.709 (0.003)
CO	-0.741 (0.012)
ID	-0.989 (0.028)
KS	-0.388 (0.013)
KY	-0.984 (0.000)
LA	-1.271 (0.000)
MA	-1.204 (0.002)
MD	-1.200 (0.000)
ME	-2.254 (0.015)
MN	-0.493 (0.000)
MO	20.000 (9.462)
NC	-0.244 (0.000)
OH	-0.712 (0.000)
OK	-0.662 (0.000)
SD	1.389 (0.072)
TN	-0.648 (0.000)
VA	0.040 (0.000)

Texas is the baseline (alpha set to 0). The other 17 alphas are recovered at the converged $\hat{\theta}$ as $\alpha_g(\hat{\theta})$ from the profile-FOC (Eq. 3 of the spec); SEs from joint Hessian sub-block.

Table 7: Four-way model comparison. AIC $= -2\log L + 2k$; BIC $= -2\log L + k\log N$. Likelihood-ratio tests reported: 8p vs 4p ($\Delta k = 4$); 8p+FE M-only vs 8p ($\Delta k = 17$); 8p+stayFE profile vs 8p ($\Delta k = 17$). All three LR statistics are far in the tail of their reference χ^2 distributions, with p-values numerically zero. The two FE specifications are non-nested to each other (same effective $k = 25$, different α-placement), so they are not LR-comparable; AIC/BIC favor the M-only joint placement by $\approx$ 1,500 units.

Model	k	N	$\log L$	AIC	BIC	LR vs nested	df	p-value
4-param	4	2,282,735	-268,481	536,971	537,021	—	—	—
8-param	8	2,282,735	-261,770	523,557	523,658	13422.2	4	0
8-param + FE (M-only, joint)	25	2,282,735	-251,613	503,277	503,593	20314.1	17	0
8-param + FE (stay, profile)	25	2,282,735	-252,365	504,779	505,095	18811.1	17	0

LR tests: 8-param vs 4-param (df=4); 8p+FE (M-only, joint) vs 8-param (df=17); 8p+FE (stay, profile) vs 8-param (df=17). Rows 3 and 4 are non-nested with each other. Both report k=25 (profile-out does not reduce free parameters of the model, only optimizer dimensionality).

Table 8: Aggregate goodness-of-fit comparison across the four specifications. Each cell reports the cell-sample-size-weighted RMSE of the model-vs-empirical residual, computed as $\sqrt{\sum_c n_c \cdot \mathrm{res}_c^2 / \sum_c n_c}$ across the four (wall$\times$regime) subgroups for the given action. Overall column collapses across all 12 wall-regime-action groups. Lower is better.

Model	Maintain	Exit	Replace	Overall
4-parameter	0.00647	0.00664	0.00369	0.00576
8-parameter	0.00345	0.00264	0.00205	0.00277
8p+FE (M-only, joint)	0.00342	0.00283	0.00193	0.00280
8p+stayFE (profile)	0.00346	0.00303	0.00200	0.00289

the wall/regime split matters most), while *Replace* fit is already comparable across the 8p variants.

5 Goodness of Fit: Model-Implied vs. Empirical Cell Shares

This section reports the model-implied conditional choice probability $\hat{P}(\text{action} \mid \text{state})$ at the converged $\hat{\theta}$ alongside the empirical action share in each (wall $\times$ regime $\times$ age bin) cell. Empirical shares are computed by direct cell aggregation on the observed panel. Model-implied shares are read from the NPL-converged $\hat{P}$ matrix saved in each fit object. For the 8p+FE specification, $\hat{P}$ is the alpha-augmented, g-marginalized measurement probability (the same object the NPL outer loop uses for its CCP update).

Each model has two figures (one per wall type), each with three facets (Maintain / Exit / Replace) and two colored lines (FF / RB regime). All six figures are produced by a single shared plotting routine so visual comparability across specifications is structural.

Table 9: 4-parameter weighted-RMSE summary of model-vs-empirical residuals by wall, regime, and action. Residual = empirical − model share; weights are cell sample size.

wall	regime	action	N_cells	total_n	unweighted_mean_residual	weighted_mean_residual	unweighted_RMSE	weighted_RMSE	max_abs_residual
DW	FF	Exit	8	351597	-0.00870	-0.00913	0.00959	0.01014	0.01795
DW	RB	Exit	8	78616	-0.01022	-0.01222	0.01125	0.01326	0.02022
SW	FF	Exit	8	1656628	0.00242	0.00237	0.00441	0.00453	0.00839
SW	RB	Exit	8	195894	-0.00055	0.00212	0.00943	0.00923	0.02020
DW	FF	Maintain	8	351597	0.00456	0.00411	0.00809	0.00872	0.01913
DW	RB	Maintain	8	78616	-0.00231	0.00351	0.01247	0.01396	0.02186
SW	FF	Maintain	8	1656628	-0.00156	-0.00106	0.00446	0.00472	0.00924
SW	RB	Maintain	8	195894	0.00170	-0.00093	0.00972	0.00926	0.01957
DW	FF	Replace	8	351597	0.00414	0.00502	0.00526	0.00616	0.00923
DW	RB	Replace	8	78616	0.01253	0.00871	0.01493	0.01238	0.02386
SW	FF	Replace	8	1656628	-0.00086	-0.00131	0.00160	0.00179	0.00225
SW	RB	Replace	8	195894	-0.00116	-0.00119	0.00127	0.00132	0.00212

5.1 4-parameter fit

Figure 1: 4-parameter fit, Single-Walled tanks. Lines are model-implied $\hat{P}(\text{action} \mid \text{state})$; open circles are empirical cell shares with area proportional to cell sample size.

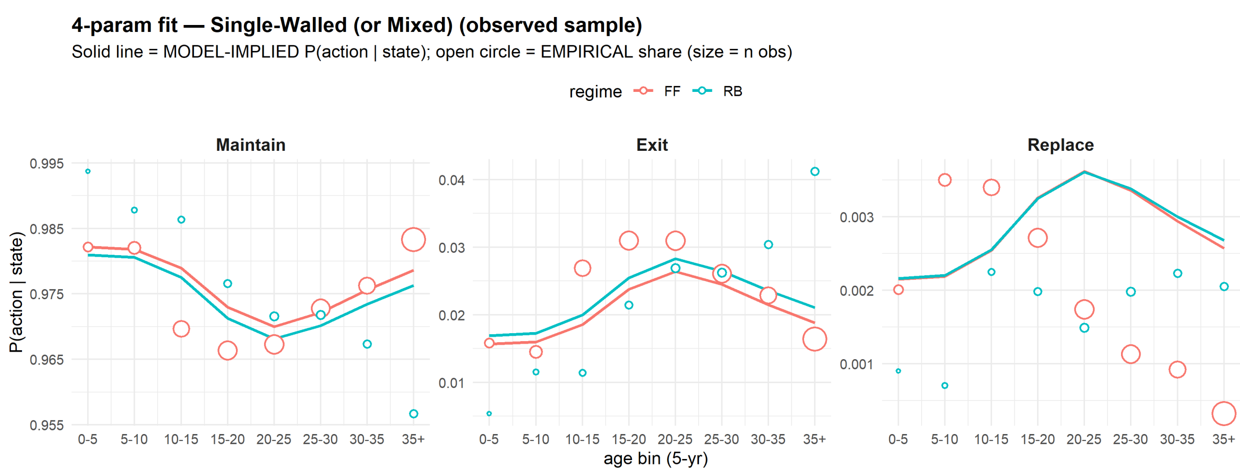
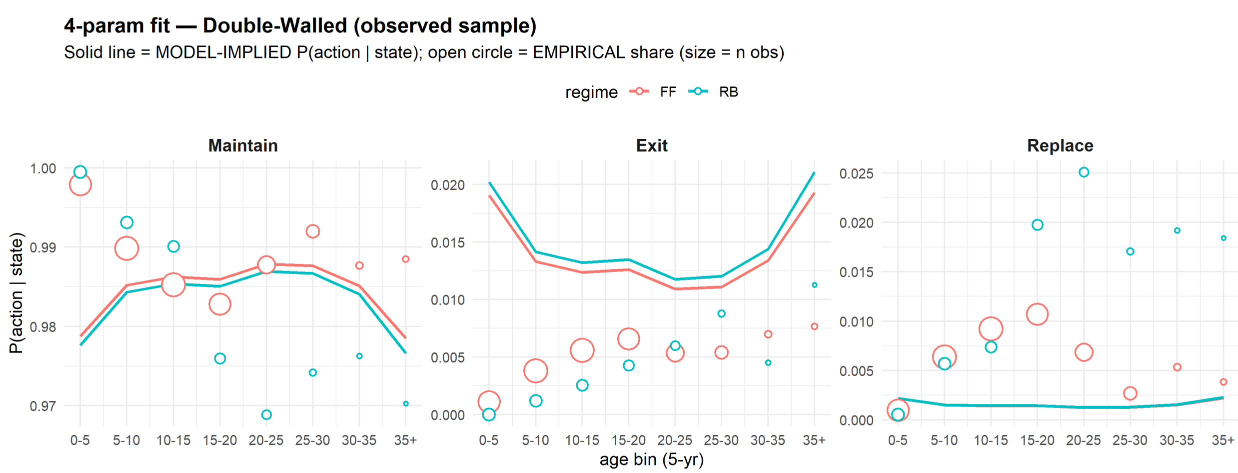


Figure 2: 4-parameter fit, Double-Walled tanks. Same construction as Figure 1.



5.2 8-parameter fit

Figure 3: 8-parameter fit, Single-Walled tanks. Same construction as Figure 1.

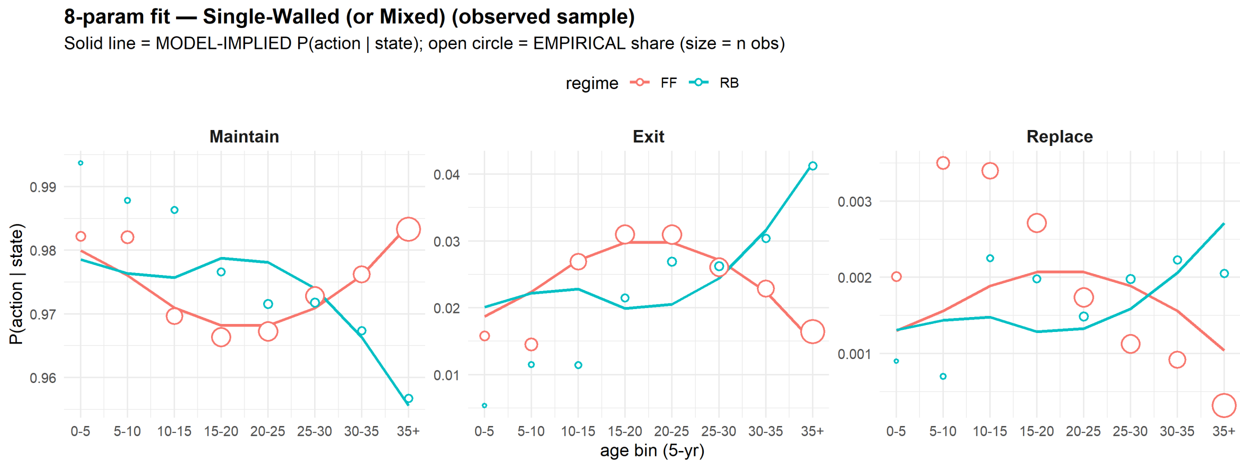


Figure 4: 8-parameter fit, Double-Walled tanks. Same construction as Figure 1.

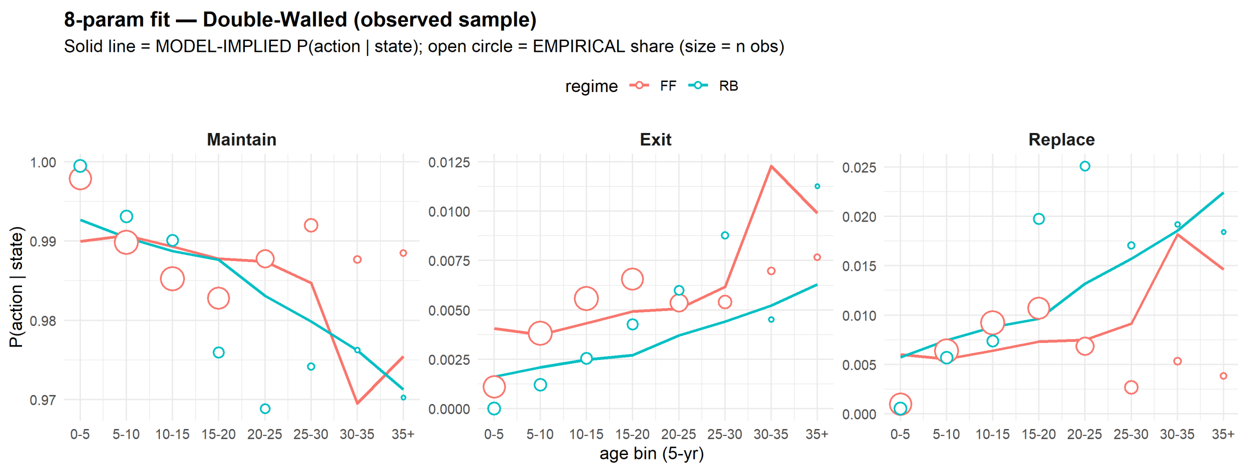


Table 10: 8-parameter weighted-RMSE summary by wall, regime, and action. Same construction as Table 9.

wall	regime	action	N_cells	total_n	weighted_RMSE	weighted_mean_residual	max_abs_residual
DW	FF	Exit	8	351597	0.00170	-0.00006	0.00532
DW	RB	Exit	8	78616	0.00176	0.00024	0.00497
SW	FF	Exit	8	1656628	0.00227	0.00005	0.00792
SW	RB	Exit	8	195894	0.00558	-0.00042	0.01475
DW	FF	Maintain	8	351597	0.00537	0.00031	0.01818
DW	RB	Maintain	8	78616	0.00773	-0.00140	0.01421
SW	FF	Maintain	8	1656628	0.00193	-0.00003	0.00598
SW	RB	Maintain	8	195894	0.00563	0.00027	0.01516
DW	FF	Replace	8	351597	0.00378	-0.00024	0.01286
DW	RB	Replace	8	78616	0.00637	0.00117	0.01192
SW	FF	Replace	8	1656628	0.00090	-0.00002	0.00194
SW	RB	Replace	8	195894	0.00051	0.00015	0.00077

5.3 8-parameter + FE (M-only, joint) fit

Figure 5: 8-parameter + FE fit, Single-Walled tanks. Model-implied probabilities marginalize the per-g choice probability over the empirical state composition $w_g(s)$. Same plot construction as Figure 1.

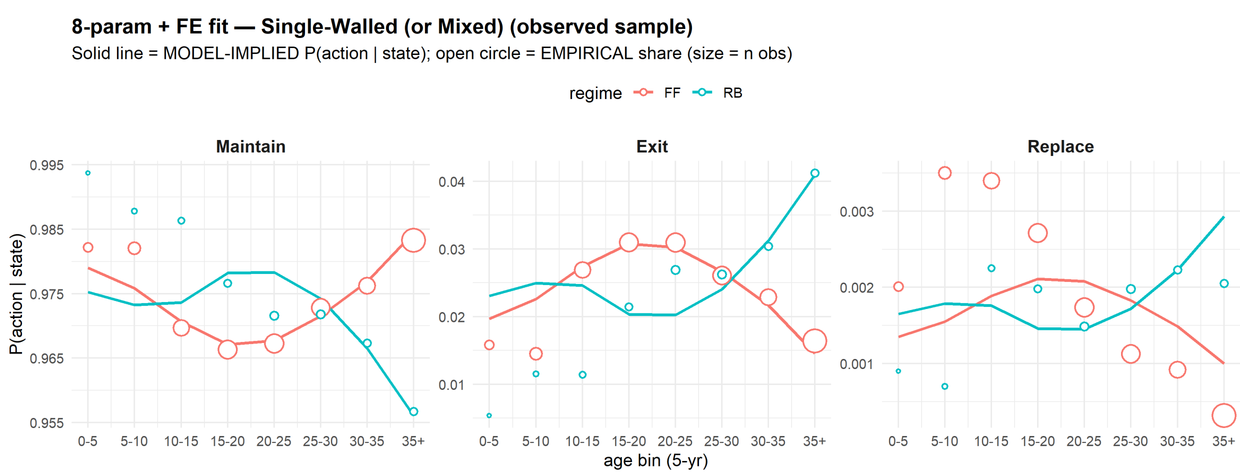
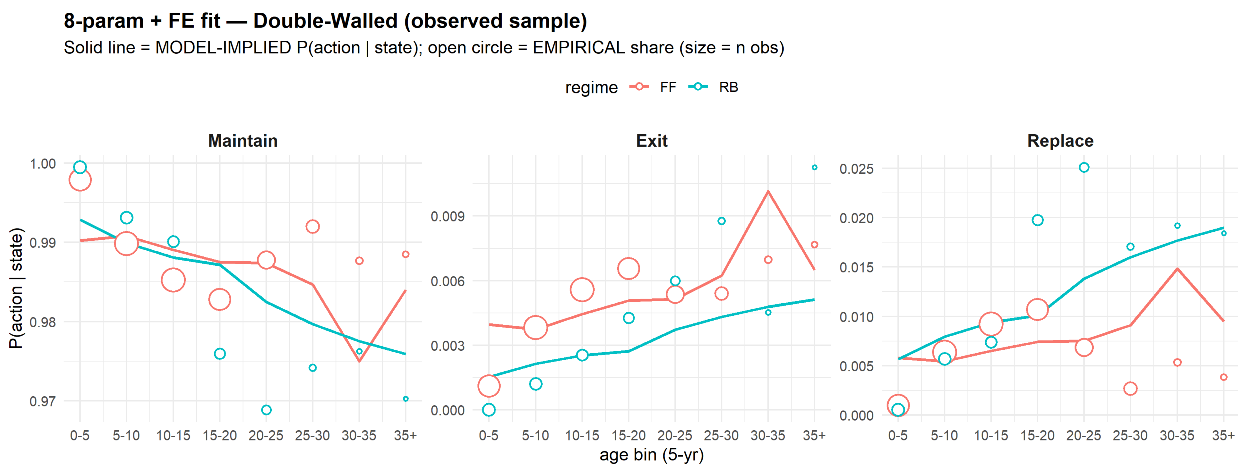


Table 11: 8-parameter + FE (M-only, joint) weighted-RMSE summary by wall, regime, and action. Same construction as Table 9.

wall	regime	action	N_cells	total_n	weighted_RMSE	weighted_mean_residual	max_abs_residual
DW	FF	Exit	8	351597	0.00154	-0.00005	0.00317
DW	RB	Exit	8	78616	0.00180	0.00026	0.00615
SW	FF	Exit	8	1656628	0.00235	0.00002	0.00810
SW	RB	Exit	8	195894	0.00642	-0.00054	0.01771
DW	FF	Maintain	8	351597	0.00486	0.00020	0.01267
DW	RB	Maintain	8	78616	0.00754	-0.00114	0.01359
SW	FF	Maintain	8	1656628	0.00184	-0.00003	0.00616
SW	RB	Maintain	8	195894	0.00652	0.00057	0.01846
DW	FF	Replace	8	351597	0.00345	-0.00015	0.00950
DW	RB	Replace	8	78616	0.00616	0.00088	0.01132
SW	FF	Replace	8	1656628	0.00088	0.00000	0.00195
SW	RB	Replace	8	195894	0.00052	-0.00003	0.00109

Figure 6: 8-parameter + FE fit, Double-Walled tanks. Same construction as Figure 5.



5.4 8-parameter + stayFE (profile) fit

Figure 7: 8-parameter + stayFE (profile) fit, Single-Walled tanks. Same construction as Figure 1. Model-implied probabilities marginalize the per-g choice probability over the empirical state composition $w_g(s)$, with α_g entering both Maintain and Replace utilities (profiled out via 1-D FOC).

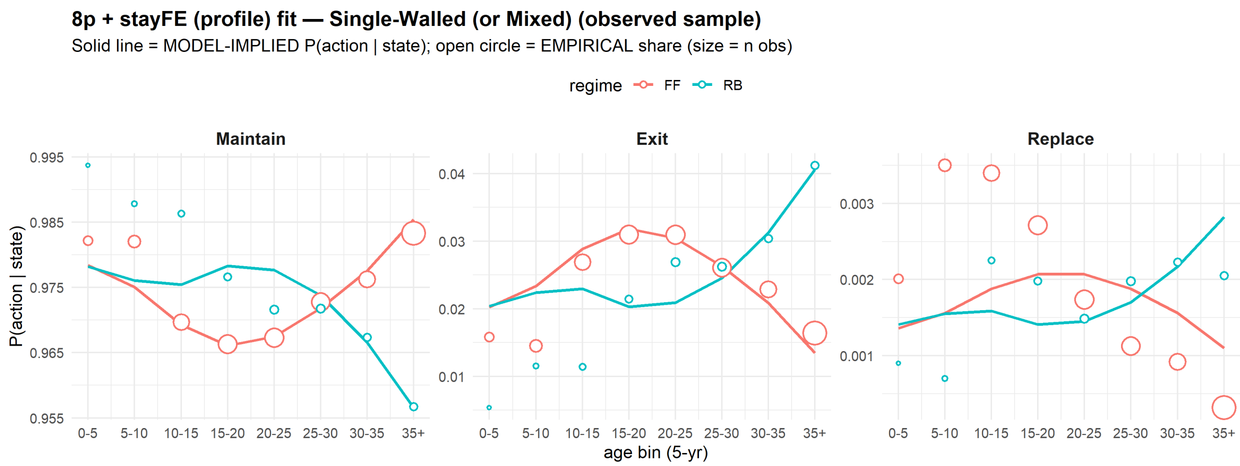
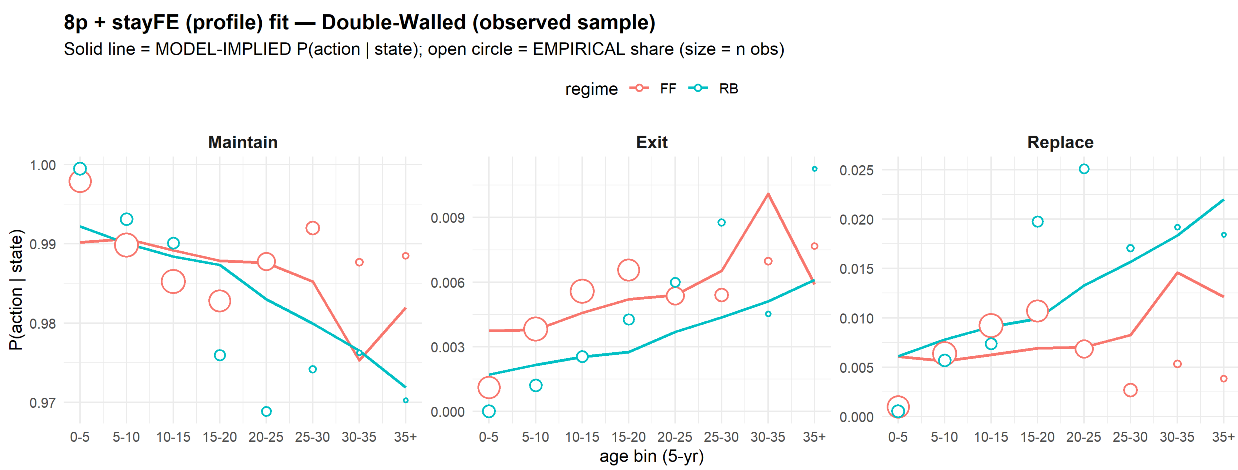


Figure 8: 8-parameter + stayFE (profile) fit, Double-Walled tanks. Same construction as Figure 7.



6 Appendix: Per-cell fit numbers

The following three tables report the full 32-row cell-level fit numbers for each specification. Columns: `Reg` (regime: FF/RB), `Wall` (SW/DW), `Age` (5-year bin), `N` (cell sample size), and for each of the three actions $a \in \{M, E, R\}$, `mod_a` $= \hat{P}_a(s)$ (model-implied), `emp_a` (empirical share), and `res_a` $=$ `emp_a` $-$ `mod_a` (residual).

6.1 4-parameter

Table 12: 8-parameter + stayFE (profile) weighted-RMSE summary by wall, regime, and action. Same construction as Table 9.

wall	regime	action	N_cells	total_n	weighted_RMSE	weighted_mean_residual	max_abs_residual
DW	FF	Exit	8	351597	0.00143	-0.00011	0.00313
DW	RB	Exit	8	78616	0.00180	0.00019	0.00516
SW	FF	Exit	8	1656628	0.00290	0.00002	0.00889
SW	RB	Exit	8	195894	0.00554	-0.00036	0.01503
DW	FF	Maintain	8	351597	0.00492	0.00011	0.01239
DW	RB	Maintain	8	78616	0.00778	-0.00109	0.01412
SW	FF	Maintain	8	1656628	0.00219	0.00001	0.00695
SW	RB	Maintain	8	195894	0.00557	0.00032	0.01555
DW	FF	Replace	8	351597	0.00359	0.00000	0.00926
DW	RB	Replace	8	78616	0.00639	0.00090	0.01182
SW	FF	Replace	8	1656628	0.00092	-0.00003	0.00194
SW	RB	Replace	8	195894	0.00049	0.00004	0.00085

Table 13: 4-parameter per-cell fit: model-implied probability, empirical share, and residual by state cell.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9788	0.9979	0.0191	0.0191	0.0011	-0.0180	0.0022	0.0010	-0.0012
FF	DW	5-10	77103	0.9852	0.9898	0.0046	0.0133	0.0038	-0.0095	0.0015	0.0064	0.0048
FF	DW	10-15	77966	0.9862	0.9852	-0.0010	0.0124	0.0056	-0.0068	0.0014	0.0092	0.0078
FF	DW	15-20	62241	0.9860	0.9828	-0.0032	0.0126	0.0066	-0.0061	0.0014	0.0107	0.0092
FF	DW	20-25	41520	0.9878	0.9878	-0.0001	0.0109	0.0054	-0.0055	0.0012	0.0069	0.0056
FF	DW	25-30	20407	0.9876	0.9920	0.0043	0.0111	0.0054	-0.0057	0.0013	0.0026	0.0014
FF	DW	30-35	4872	0.9851	0.9877	0.0026	0.0134	0.0070	-0.0064	0.0015	0.0053	0.0038
FF	DW	35+	2867	0.9785	0.9885	0.0100	0.0193	0.0077	-0.0116	0.0022	0.0038	0.0016
FF	SW	0-5	46709	0.9822	0.9822	0.0000	0.0157	0.0158	0.0001	0.0022	0.0020	-0.0001
FF	SW	5-10	97999	0.9818	0.9820	0.0002	0.0160	0.0145	-0.0015	0.0022	0.0035	0.0013
FF	SW	10-15	180163	0.9789	0.9697	-0.0092	0.0185	0.0269	0.0084	0.0025	0.0034	0.0009
FF	SW	15-20	240344	0.9730	0.9663	-0.0066	0.0238	0.0310	0.0072	0.0033	0.0027	-0.0006
FF	SW	20-25	255278	0.9700	0.9673	-0.0027	0.0264	0.0310	0.0046	0.0036	0.0017	-0.0019
FF	SW	25-30	234030	0.9722	0.9728	0.0006	0.0245	0.0261	0.0016	0.0034	0.0011	-0.0022
FF	SW	30-35	191377	0.9756	0.9762	0.0006	0.0215	0.0228	0.0014	0.0029	0.0009	-0.0020
FF	SW	35+	410728	0.9786	0.9833	0.0046	0.0188	0.0164	-0.0024	0.0026	0.0003	-0.0022
RB	DW	0-5	18813	0.9776	0.9995	0.0219	0.0202	0.0000	-0.0202	0.0022	0.0005	-0.0016
RB	DW	5-10	16557	0.9843	0.9931	0.0088	0.0142	0.0012	-0.0130	0.0015	0.0057	0.0042
RB	DW	10-15	14499	0.9854	0.9901	0.0047	0.0132	0.0026	-0.0106	0.0014	0.0074	0.0060
RB	DW	15-20	13106	0.9851	0.9760	-0.0091	0.0135	0.0043	-0.0092	0.0014	0.0198	0.0183
RB	DW	20-25	8677	0.9870	0.9689	-0.0181	0.0118	0.0060	-0.0058	0.0013	0.0251	0.0239
RB	DW	25-30	4218	0.9867	0.9742	-0.0125	0.0120	0.0088	-0.0033	0.0013	0.0171	0.0158
RB	DW	30-35	1770	0.9841	0.9763	-0.0078	0.0144	0.0045	-0.0099	0.0015	0.0192	0.0177
RB	DW	35+	976	0.9766	0.9703	-0.0063	0.0211	0.0113	-0.0098	0.0023	0.0184	0.0162
RB	SW	0-5	3350	0.9810	0.9937	0.0128	0.0169	0.0054	-0.0115	0.0022	0.0009	-0.0013
RB	SW	5-10	9999	0.9806	0.9878	0.0072	0.0172	0.0115	-0.0057	0.0022	0.0007	-0.0015
RB	SW	10-15	19093	0.9775	0.9863	0.0088	0.0200	0.0114	-0.0085	0.0026	0.0022	-0.0003
RB	SW	15-20	26212	0.9713	0.9766	0.0053	0.0255	0.0214	-0.0040	0.0032	0.0020	-0.0013
RB	SW	20-25	36841	0.9681	0.9716	0.0035	0.0283	0.0269	-0.0014	0.0036	0.0015	-0.0021
RB	SW	25-30	37789	0.9702	0.9718	0.0016	0.0265	0.0262	-0.0003	0.0034	0.0020	-0.0014

RB	SW	30-35	31888	0.9734	0.9674	-0.0061	0.0236	0.0304	0.0069	0.0030	0.0022	-0.0008
RB	SW	35+	30722	0.9763	0.9567	-0.0196	0.0210	0.0412	0.0202	0.0027	0.0021	-0.0006

6.2 8-parameter

Table 14: 8-parameter per-cell fit. Same construction as Table 13.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9899	0.9979	0.0080	0.0041	0.0011	-0.0029	0.0060	0.0010	-0.0050
FF	DW	5-10	77103	0.9907	0.9898	-0.0009	0.0037	0.0038	0.0001	0.0056	0.0064	0.0008
FF	DW	10-15	77966	0.9893	0.9852	-0.0040	0.0043	0.0056	0.0013	0.0064	0.0092	0.0028
FF	DW	15-20	62241	0.9878	0.9828	-0.0050	0.0049	0.0066	0.0016	0.0073	0.0107	0.0034
FF	DW	20-25	41520	0.9874	0.9878	0.0003	0.0051	0.0054	0.0003	0.0075	0.0069	-0.0006
FF	DW	25-30	20407	0.9847	0.9920	0.0073	0.0062	0.0054	-0.0008	0.0091	0.0026	-0.0065
FF	DW	30-35	4872	0.9695	0.9877	0.0182	0.0123	0.0070	-0.0053	0.0182	0.0053	-0.0129
FF	DW	35+	2867	0.9755	0.9885	0.0130	0.0099	0.0077	-0.0022	0.0146	0.0038	-0.0108
FF	SW	0-5	46709	0.9800	0.9822	0.0022	0.0187	0.0158	-0.0029	0.0013	0.0020	0.0007
FF	SW	5-10	97999	0.9760	0.9820	0.0060	0.0224	0.0145	-0.0079	0.0016	0.0035	0.0019
FF	SW	10-15	180163	0.9710	0.9697	-0.0013	0.0271	0.0269	-0.0002	0.0019	0.0034	0.0015
FF	SW	15-20	240344	0.9682	0.9663	-0.0019	0.0297	0.0310	0.0012	0.0021	0.0027	0.0006
FF	SW	20-25	255278	0.9682	0.9673	-0.0009	0.0298	0.0310	0.0012	0.0021	0.0017	-0.0003
FF	SW	25-30	234030	0.9709	0.9728	0.0019	0.0272	0.0261	-0.0011	0.0019	0.0011	-0.0008
FF	SW	30-35	191377	0.9761	0.9762	0.0002	0.0224	0.0228	0.0005	0.0016	0.0009	-0.0006
FF	SW	35+	410728	0.9840	0.9833	-0.0007	0.0150	0.0164	0.0014	0.0010	0.0003	-0.0007
RB	DW	0-5	18813	0.9927	0.9995	0.0068	0.0016	0.0000	-0.0016	0.0057	0.0005	-0.0052
RB	DW	5-10	16557	0.9905	0.9931	0.0027	0.0021	0.0012	-0.0009	0.0074	0.0057	-0.0018
RB	DW	10-15	14499	0.9888	0.9901	0.0013	0.0025	0.0026	0.0001	0.0088	0.0074	-0.0014
RB	DW	15-20	13106	0.9877	0.9760	-0.0117	0.0027	0.0043	0.0016	0.0096	0.0198	0.0101
RB	DW	20-25	8677	0.9831	0.9689	-0.0142	0.0037	0.0060	0.0023	0.0132	0.0251	0.0119
RB	DW	25-30	4218	0.9798	0.9742	-0.0057	0.0044	0.0088	0.0043	0.0157	0.0171	0.0013
RB	DW	30-35	1770	0.9762	0.9763	0.0001	0.0052	0.0045	-0.0007	0.0186	0.0192	0.0006
RB	DW	35+	976	0.9713	0.9703	-0.0010	0.0063	0.0113	0.0050	0.0224	0.0184	-0.0040
RB	SW	0-5	3350	0.9786	0.9937	0.0152	0.0201	0.0054	-0.0147	0.0013	0.0009	-0.0004
RB	SW	5-10	9999	0.9764	0.9878	0.0114	0.0222	0.0115	-0.0107	0.0014	0.0007	-0.0007
RB	SW	10-15	19093	0.9757	0.9863	0.0106	0.0228	0.0114	-0.0114	0.0015	0.0022	0.0008
RB	SW	15-20	26212	0.9788	0.9766	-0.0022	0.0199	0.0214	0.0015	0.0013	0.0020	0.0007
RB	SW	20-25	36841	0.9781	0.9716	-0.0065	0.0205	0.0269	0.0064	0.0013	0.0015	0.0002
RB	SW	25-30	37789	0.9740	0.9718	-0.0022	0.0244	0.0262	0.0018	0.0016	0.0020	0.0004
RB	SW	30-35	31888	0.9663	0.9674	0.0010	0.0316	0.0304	-0.0012	0.0021	0.0022	0.0002
RB	SW	35+	30722	0.9556	0.9567	0.0011	0.0417	0.0412	-0.0004	0.0027	0.0021	-0.0007

6.3 8-parameter + FE (M-only, joint)

Table 15: 8-parameter + FE (M-only, joint) per-cell fit. Same construction as Table 13.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9902	0.9979	0.0076	0.0040	0.0011	-0.0029	0.0058	0.0010	-0.0048
FF	DW	5-10	77103	0.9908	0.9898	-0.0010	0.0037	0.0038	0.0001	0.0055	0.0064	0.0009
FF	DW	10-15	77966	0.9890	0.9852	-0.0038	0.0044	0.0056	0.0011	0.0065	0.0092	0.0027

FF	DW	15-20	62241	0.9875	0.9828	-0.0047	0.0051	0.0066	0.0015	0.0074	0.0107	0.0032
FF	DW	20-25	41520	0.9874	0.9878	0.0004	0.0051	0.0054	0.0002	0.0075	0.0069	-0.0006
FF	DW	25-30	20407	0.9846	0.9920	0.0073	0.0062	0.0054	-0.0008	0.0091	0.0026	-0.0065
FF	DW	30-35	4872	0.9750	0.9877	0.0127	0.0101	0.0070	-0.0032	0.0148	0.0053	-0.0095
FF	DW	35+	2867	0.9840	0.9885	0.0045	0.0065	0.0077	0.0012	0.0095	0.0038	-0.0056
FF	SW	0-5	46709	0.9790	0.9822	0.0032	0.0197	0.0158	-0.0039	0.0014	0.0020	0.0007
FF	SW	5-10	97999	0.9759	0.9820	0.0062	0.0226	0.0145	-0.0081	0.0015	0.0035	0.0019
FF	SW	10-15	180163	0.9707	0.9697	-0.0010	0.0274	0.0269	-0.0005	0.0019	0.0034	0.0015
FF	SW	15-20	240344	0.9671	0.9663	-0.0008	0.0307	0.0310	0.0002	0.0021	0.0027	0.0006
FF	SW	20-25	255278	0.9677	0.9673	-0.0004	0.0302	0.0310	0.0008	0.0021	0.0017	-0.0003
FF	SW	25-30	234030	0.9715	0.9728	0.0013	0.0266	0.0261	-0.0006	0.0018	0.0011	-0.0007
FF	SW	30-35	191377	0.9769	0.9762	-0.0006	0.0217	0.0228	0.0012	0.0015	0.0009	-0.0006
FF	SW	35+	410728	0.9845	0.9833	-0.0012	0.0145	0.0164	0.0019	0.0010	0.0003	-0.0007
RB	DW	0-5	18813	0.9928	0.9995	0.0066	0.0015	0.0000	-0.0015	0.0056	0.0005	-0.0051
RB	DW	5-10	16557	0.9899	0.9931	0.0032	0.0021	0.0012	-0.0009	0.0079	0.0057	-0.0023
RB	DW	10-15	14499	0.9881	0.9901	0.0020	0.0025	0.0026	0.0000	0.0094	0.0074	-0.0020
RB	DW	15-20	13106	0.9872	0.9760	-0.0112	0.0027	0.0043	0.0015	0.0101	0.0198	0.0097
RB	DW	20-25	8677	0.9825	0.9689	-0.0136	0.0037	0.0060	0.0023	0.0138	0.0251	0.0113
RB	DW	25-30	4218	0.9797	0.9742	-0.0055	0.0043	0.0088	0.0045	0.0160	0.0171	0.0011
RB	DW	30-35	1770	0.9775	0.9763	-0.0013	0.0048	0.0045	-0.0003	0.0177	0.0192	0.0015
RB	DW	35+	976	0.9759	0.9703	-0.0056	0.0051	0.0113	0.0062	0.0190	0.0184	-0.0005
RB	SW	0-5	3350	0.9753	0.9937	0.0185	0.0231	0.0054	-0.0177	0.0016	0.0009	-0.0008
RB	SW	5-10	9999	0.9732	0.9878	0.0146	0.0250	0.0115	-0.0135	0.0018	0.0007	-0.0011
RB	SW	10-15	19093	0.9736	0.9863	0.0127	0.0246	0.0114	-0.0132	0.0018	0.0022	0.0005
RB	SW	15-20	26212	0.9782	0.9766	-0.0016	0.0204	0.0214	0.0011	0.0015	0.0020	0.0005
RB	SW	20-25	36841	0.9783	0.9716	-0.0067	0.0202	0.0269	0.0067	0.0014	0.0015	0.0000
RB	SW	25-30	37789	0.9742	0.9718	-0.0024	0.0240	0.0262	0.0022	0.0017	0.0020	0.0003
RB	SW	30-35	31888	0.9665	0.9674	0.0008	0.0312	0.0304	-0.0008	0.0022	0.0022	0.0000
RB	SW	35+	30722	0.9561	0.9567	0.0006	0.0410	0.0412	0.0003	0.0029	0.0021	-0.0009

6.4 8-parameter + stayFE (profile)

Table 16: 8-parameter + stayFE (profile) per-cell fit. Same construction as Table 13.

Reg	Wall	Age	N	mod_M	emp_M	res_M	mod_E	emp_E	res_E	mod_R	emp_R	res_R
FF	DW	0-5	64621	0.9902	0.9979	0.0077	0.0037	0.0011	-0.0026	0.0061	0.0010	-0.0051
FF	DW	5-10	77103	0.9906	0.9898	-0.0008	0.0038	0.0038	0.0000	0.0056	0.0064	0.0007
FF	DW	10-15	77966	0.9892	0.9852	-0.0039	0.0046	0.0056	0.0010	0.0062	0.0092	0.0030
FF	DW	15-20	62241	0.9879	0.9828	-0.0051	0.0052	0.0066	0.0014	0.0069	0.0107	0.0037
FF	DW	20-25	41520	0.9876	0.9878	0.0002	0.0054	0.0054	0.0000	0.0070	0.0069	-0.0002
FF	DW	25-30	20407	0.9852	0.9920	0.0067	0.0065	0.0054	-0.0011	0.0082	0.0026	-0.0056
FF	DW	30-35	4872	0.9753	0.9877	0.0124	0.0101	0.0070	-0.0031	0.0146	0.0053	-0.0093
FF	DW	35+	2867	0.9820	0.9885	0.0065	0.0059	0.0077	0.0018	0.0121	0.0038	-0.0083
FF	SW	0-5	46709	0.9784	0.9822	0.0038	0.0202	0.0158	-0.0044	0.0014	0.0020	0.0007
FF	SW	5-10	97999	0.9751	0.9820	0.0069	0.0234	0.0145	-0.0089	0.0016	0.0035	0.0019
FF	SW	10-15	180163	0.9692	0.9697	0.0004	0.0289	0.0269	-0.0019	0.0019	0.0034	0.0015
FF	SW	15-20	240344	0.9661	0.9663	0.0002	0.0318	0.0310	-0.0009	0.0021	0.0027	0.0006
FF	SW	20-25	255278	0.9675	0.9673	-0.0002	0.0305	0.0310	0.0005	0.0021	0.0017	-0.0003
FF	SW	25-30	234030	0.9718	0.9728	0.0010	0.0263	0.0261	-0.0002	0.0019	0.0011	-0.0008
FF	SW	30-35	191377	0.9776	0.9762	-0.0013	0.0209	0.0228	0.0020	0.0016	0.0009	-0.0006

FF	SW	35+	410728	0.9854	0.9833	-0.0022	0.0135	0.0164	0.0030	0.0011	0.0003	-0.0008
RB	DW	0-5	18813	0.9922	0.9995	0.0073	0.0017	0.0000	-0.0017	0.0061	0.0005	-0.0056
RB	DW	5-10	16557	0.9900	0.9931	0.0031	0.0022	0.0012	-0.0010	0.0078	0.0057	-0.0021
RB	DW	10-15	14499	0.9884	0.9901	0.0017	0.0025	0.0026	0.0000	0.0091	0.0074	-0.0017
RB	DW	15-20	13106	0.9873	0.9760	-0.0114	0.0027	0.0043	0.0015	0.0099	0.0198	0.0098
RB	DW	20-25	8677	0.9830	0.9689	-0.0141	0.0037	0.0060	0.0023	0.0133	0.0251	0.0118
RB	DW	25-30	4218	0.9800	0.9742	-0.0058	0.0044	0.0088	0.0044	0.0157	0.0171	0.0014
RB	DW	30-35	1770	0.9765	0.9763	-0.0003	0.0051	0.0045	-0.0006	0.0184	0.0192	0.0008
RB	DW	35+	976	0.9719	0.9703	-0.0016	0.0061	0.0113	0.0052	0.0220	0.0184	-0.0036
RB	SW	0-5	3350	0.9782	0.9937	0.0156	0.0204	0.0054	-0.0150	0.0014	0.0009	-0.0005
RB	SW	5-10	9999	0.9761	0.9878	0.0117	0.0224	0.0115	-0.0109	0.0015	0.0007	-0.0008
RB	SW	10-15	19093	0.9754	0.9863	0.0109	0.0230	0.0114	-0.0116	0.0016	0.0022	0.0007
RB	SW	15-20	26212	0.9783	0.9766	-0.0018	0.0203	0.0214	0.0012	0.0014	0.0020	0.0006
RB	SW	20-25	36841	0.9777	0.9716	-0.0061	0.0209	0.0269	0.0060	0.0014	0.0015	0.0000
RB	SW	25-30	37789	0.9738	0.9718	-0.0020	0.0245	0.0262	0.0017	0.0017	0.0020	0.0003
RB	SW	30-35	31888	0.9666	0.9674	0.0008	0.0313	0.0304	-0.0009	0.0022	0.0022	0.0001
RB	SW	35+	30722	0.9566	0.9567	0.0001	0.0406	0.0412	0.0006	0.0028	0.0021	-0.0008